G - Copy Query

分值：600 分

Problem Statement

There are N sequences of length M: $A_1, A_2, \dots, A_N$. Initially, all elements of all sequences are 0. Hereafter, the j-th element of sequence A_i is denoted by $A_{i,j}$.

共有 N 个长度为 M 的序列：$A_1, A_2, \dots, A_N$。初始时，所有序列的所有元素均为 0。 以下，序列 A_i 的第 j 个元素记为 $A_{i,j}$。

Process the following three types of queries in the given order, Q queries in total.

按给定顺序处理以下三种类型的查询，总共有 Q 个查询。

- Type 1: Overwrite sequence A_{X_i} with sequence A_{Y_i}. That is, for every integer j ($1 \leq j \leq M$), change $A_{X_i,j}$ to $A_{Y_i,j}$.
- Type 2: Change the Y_i-th element of sequence A_{X_i}, that is, A_{X_i,Y_i}, to Z_i.
- Type 3: For sequence A_{X_i}, output the sum of elements from the L_i-th through the R_i-th, that is, $A_{X_i,L_i} + A_{X_i,L_i+1} + \cdots + A_{X_i,R_i}$.
- 类型 1：用序列 A_{Y_i} 覆盖序列 A_{X_i}。即对于每个整数 j（$1 \leq j \leq M$），将 $A_{X_i,j}$ 修改为 $A_{Y_i,j}$。
- 类型 2：修改序列 A_{X_i} 的第 Y_i 个元素，即将 A_{X_i,Y_i} 修改为 Z_i。
- 类型 3：对于序列 A_{X_i}，输出第 L_i 个到第 R_i 个元素的和，即 $A_{X_i,L_i} + A_{X_i,L_i+1} + \cdots + A_{X_i,R_i}$。

Constraints

- $1 \leq N, M \leq 2 \times 10^5$
- $1 \leq Q \leq 2 \times 10^5$
- Type 1 queries satisfy the following constraints:
  - $1 \leq X_i, Y_i \leq N$
- Type 2 queries satisfy the following constraints:

- $1 \leq X_i \leq N$
- $1 \leq Y_i \leq M$
- $0 \leq Z_i \leq 10^9$

- Type 3 queries satisfy the following constraints:
  - $1 \leq X_i \leq N$
  - $1 \leq L_i \leq R_i \leq M$
- All input values are integers.
- $1 \leq N, M \leq 2 \times 10^5$
- $1 \leq Q \leq 2 \times 10^5$
- 类型 1 的查询满足以下约束条件：
  - $1 \leq X_i, Y_i \leq N$
- 类型 2 的查询满足以下约束条件：
  - $1 \leq X_i \leq N$
  - $1 \leq Y_i \leq M$
  - $0 \leq Z_i \leq 10^9$
- 类型 3 的查询满足以下约束条件：
  - $1 \leq X_i \leq N$
  - $1 \leq L_i \leq R_i \leq M$
- 所有输入值均为整数。

Input

The input is given from Standard Input in the following format:

输入按照以下格式从标准输入给出：

$$
\begin{aligned}
&N\ M\ Q \\
&\mathrm{Query}_1 \\
&\mathrm{Query}_2 \\
&\vdots \\
&\mathrm{Query}_Q
\end{aligned}
$$

Here, Query_i represents the i-th query.

其中，Query_i 表示第 i 个查询。

A type 1 query is given in the following format:

类型 1 的查询按照以下格式给出：

$1\ X_i\ Y_i$

A type 2 query is given in the following format:

类型 2 的查询按照以下格式给出：

$2\ X_i\ Y_i\ Z_i$

A type 3 query is given in the following format:

类型 3 的查询按照以下格式给出：

$3\ X_i\ L_i\ R_i$

Output

For each type 3 query, output the answer to that query. If there are no type 3 queries, the output should be empty.

对于每个类型 3 的查询，输出该查询的答案。如果没有类型 3 的查询，则输出为空。

Sample Input 1

```
4 5 10
2 2 1 2
2 2 2 7
2 2 4 8
1 1 2
2 2 3 1
1 3 2
2 3 2 10
3 1 2 4
3 2 1 4
3 3 2 2
```

Sample Output 1

```
15
18
10
```

In this input, prepare four sequences of length 5.

在该输入中，准备四个长度为 5 的序列。

- In the 1st query, set $A_{2,1} = 2$. At this point, $A_2 = (2, 0, 0, 0, 0)$.
- In the 2nd query, set $A_{2,2} = 7$. At this point, $A_2 = (2, 7, 0, 0, 0)$.
- In the 3rd query, set $A_{2,4} = 8$. At this point, $A_2 = (2, 7, 0, 8, 0)$.
- In the 4th query, overwrite sequence A_1 with sequence A_2. At this point, $A_1 = (2, 7, 0, 8, 0)$.
- In the 5th query, set $A_{2,3} = 1$. At this point, $A_2 = (2, 7, 1, 8, 0)$.
- In the 6th query, overwrite sequence A_3 with sequence A_2. At this point, $A_3 = (2, 7, 1, 8, 0)$.
- In the 7th query, set $A_{3,2} = 10$. At this point, $A_3 = (2, 10, 1, 8, 0)$.
- In the 8th query, output $A_{1,2} + A_{1,3} + A_{1,4} = 7 + 0 + 8 = 15$.
- In the 9th query, output $A_{2,1} + A_{2,2} + A_{2,3} + A_{2,4} = 2 + 7 + 1 + 8 = 18$.
- In the 10th query, output $A_{3,2} = 10$.
- 第 1 个查询中，设置 $A_{2,1} = 2$。此时 $A_2 = (2, 0, 0, 0, 0)$。
- 第 2 个查询中，设置 $A_{2,2} = 7$。此时 $A_2 = (2, 7, 0, 0, 0)$。
- 第 3 个查询中，设置 $A_{2,4} = 8$。此时 $A_2 = (2, 7, 0, 8, 0)$。
- 第 4 个查询中，用序列 A_2 覆盖序列 A_1。此时 $A_1 = (2, 7, 0, 8, 0)$。
- 第 5 个查询中，设置 $A_{2,3} = 1$。此时 $A_2 = (2, 7, 1, 8, 0)$。
- 第 6 个查询中，用序列 A_2 覆盖序列 A_3。此时 $A_3 = (2, 7, 1, 8, 0)$。
- 第 7 个查询中，设置 $A_{3,2} = 10$。此时 $A_3 = (2, 10, 1, 8, 0)$。
- 第 8 个查询中，输出 $A_{1,2} + A_{1,3} + A_{1,4} = 7 + 0 + 8 = 15$。

- 第 9 个查询中，输出 $A_{2,1}+A_{2,2}+A_{2,3}+A_{2,4}=2+7+1+8=18$。
- 第 10 个查询中，输出 $A_{3,2}=10$。